

30/07/2026

DATA COLLECTION AND FRAME WORK

1:LOGGER

The screenshot displays the SIMATS Online Programming Lab interface. The browser tabs include 'java/CO-1 ASSESSMENT TOOL', 'SIMATS Online Lab', 'ChatGPT', and 'Ask Gemini'. The address bar shows 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The lab header features the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', and the user 'Jalla Yoganjulu Reddy'. The left sidebar shows 'QUESTIONS 8 / 8 completed' with a dropdown menu set to 'Logger'. The 'LOGGER' section describes a project requiring a logging framework. The 'PROGRAM EDITOR' shows the following Java code:

```
1 interface Logger {
2     void debug();
3     void info();
4 }
5
6 class Log implements Logger {
7
8     public void debug() {
9         System.out.println("Debug Message");
10    }
11
12    public void info() {
13        System.out.println("Info Message");
14    }
15
16    public static void main(String[] args) {
17        Log l = new Log();
18        l.debug();
19        l.info();
20    }
21 }
```

The 'OUTPUT' section shows the execution results:

```
Debug Message
Info Message
```

The 'INPUT' section on the right is empty, with the text 'Your INPUT goes here!'.

2:CHARACTER

The screenshot displays the SIMATS Online Programming Lab interface for the 'Character' exercise. The browser tabs and address bar are the same as in the previous screenshot. The left sidebar shows 'QUESTIONS 8 / 8 completed' with a dropdown menu set to 'Character'. The 'CHARACTER' section describes a game development project. The 'PROGRAM EDITOR' shows the following Java code:

```
1 interface Character {
2     void move();
3     void attack();
4     void damage();
5 }
6
7 class Hero implements Character {
8
9     public void move() {
10        System.out.println("Hero moves");
11    }
12
13    public void attack() {
14        System.out.println("Hero attacks");
15    }
16
17    public void damage() {
18        System.out.println("Hero takes damage");
19    }
20
21    public static void main(String[] args) {
22
23    }
24 }
```

The 'OUTPUT' section shows the execution results:

```
Hero moves
Hero attacks
Hero takes damage
```

The 'INPUT' section on the right is empty, with the text 'Your INPUT goes here!'.

3: SORT

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'java/CD-1 ASSESSMENT TOOL', 'SIMATS Online Lab', 'ChatGPT', and 'Ask Gemini'. The URL is 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The page header displays the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', and the user 'Saila Yoganjulu Reddy'. The left sidebar has a 'QUESTIONS' section with 'Sort' selected and a description: 'Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.' The 'PROGRAM EDITOR' shows the following Java code:

```
1 import java.util.*;
2
3 public class Sortlist {
4     public static void main(String[] args) {
5         ArrayList<Integer> list = new ArrayList<>();
6
7         list.add(10);
8         list.add(40);
9         list.add(50);
10        list.add(20);
11
12        Collections.sort(list);
13
14        System.out.println(list);
15    }
16 }
```

The 'OUTPUT' section displays the result: '[10, 20, 30, 50]'. The 'INPUT' section on the right contains the text 'Your INPUT goes here!'.

4. HASHSET

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'java/CD-1 ASSESSMENT TOOL', 'SIMATS Online Lab', 'ChatGPT', and 'Ask Gemini'. The URL is 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The page header displays the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', and the user 'Saila Yoganjulu Reddy'. The left sidebar has a 'QUESTIONS' section with 'HashSet' selected and a description: 'Write a Java program to create a HashSet of strings and remove a specific element from it.' The 'PROGRAM EDITOR' shows the following Java code:

```
1 import java.util.HashSet;
2
3 public class Solution {
4     public static void main(String[] args) {
5         HashSet<String> set = new HashSet<>();
6
7         set.add("Apple");
8         set.add("Banana");
9         set.add("Orange");
10        set.add("Mango");
11
12        System.out.println(set);
13    }
14 }
```

The 'OUTPUT' section displays the result: '[Apple, Mango, Orange, Banana]'. The 'INPUT' section on the right contains the text 'Your INPUT goes here!'. A 'Snipping Tool' notification is visible in the bottom right corner, stating 'Screenshot copied to clipboard' and 'Automatically saved to screenshots folder.' with a 'Markup and share' button.

5.TREE MAP

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'java/CO-1 ASSESSMENT TOOL', 'SIMATS Online Lab', and 'ChatGPT'. The address bar shows 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The page header features the SIMATS ENGINEERING logo and 'SIMATS Online Programming Lab JAVA PROGRAMMING'. A user profile for 'Jalla Yoganjulu Reddy' is visible in the top right.

QUESTIONS 8 / 8 completed

TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

```
1 import java.util.*;
2
3 public class TreeMapDemo {
4     public static void main(String[] args) {
5         TreeMap<String, Integer> map = new TreeMap<>();
6
7         map.put("Ram", 90);
8         map.put("John", 85);
9         map.put("Ravi", 95);
10
11         for (Map.Entry<String, Integer> e : map.entrySet()) {
12             System.out.println(e.getKey() + " " + e.getValue());
13         }
14     }
15 }
```

OUTPUT

```
John 85
Ram 90
Ravi 95
```

INPUT

Your INPUT goes here!

6.PRIORITY QUEUE

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'java/CO-1 ASSESSMENT TOOL', 'SIMATS Online Lab', and 'ChatGPT'. The address bar shows 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The page header features the SIMATS ENGINEERING logo and 'SIMATS Online Programming Lab JAVA PROGRAMMING'. A user profile for 'Jalla Yoganjulu Reddy' is visible in the top right.

QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

```
1 import java.util.PriorityQueue;
2
3 public class Main {
4     public static void main(String[] args) {
5         PriorityQueue<Integer> pq = new PriorityQueue<>();
6
7         pq.add(10);
8         pq.add(10);
9         pq.add(20);
10
11         System.out.println(pq);
12     }
13 }
```

OUTPUT

```
[10, 30, 20]
```

INPUT

Your INPUT goes here!

7.LINKED HASHMAP

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'java/CO-1 ASSESSMENT TOOL', 'SIMATS Online Lab', 'ChatGPT', and 'Ask Gemini'. The URL is 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The page header shows 'SIMATS ENGINEERING' and 'SIMATS Online Programming Lab JAVA PROGRAMMING'. The user is 'Saila Yoganjulu Reddy' with a 'RESULTS SCORED' status.

QUESTIONS 8 / 8 completed

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR Java Run Save

```
1 import java.util.LinkedHashMap;
2
3 public class Main {
4     public static void main(String[] args) {
5         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
6
7         map.put("Apple", 10);
8         map.put("Banana", 20);
9         map.put("Orange", 30);
10
11         System.out.println(map.keySet());
12     }
13 }
```

OUTPUT

```
[Apple, Banana, Orange]
```

INPUT

Your INPUT goes here!

8.MAX MIN

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'java/CO-1 ASSESSMENT TOOL', 'SIMATS Online Lab', 'ChatGPT', and 'Ask Gemini'. The URL is 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections'. The page header shows 'SIMATS ENGINEERING' and 'SIMATS Online Programming Lab JAVA PROGRAMMING'. The user is 'Saila Yoganjulu Reddy' with a 'RESULTS SCORED' status.

QUESTIONS 8 / 8 completed

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR Java Run Save

```
1 import java.util.LinkedHashMap;
2
3 public class Main {
4     public static void main(String[] args) {
5         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
6
7         map.put("Apple", 10);
8         map.put("Banana", 20);
9         map.put("Orange", 30);
10
11         System.out.println(map.keySet());
12     }
13 }
```

OUTPUT

```
[Apple, Banana, Orange]
```

INPUT

Your INPUT goes here!